

THE UNIVERSITY OF AZAD JAMMU & KASHMIR
MUZAFFARABAD

DATABASE MANAGEMENT SYSTEM LAB

Lab 4 — Normalization — First Normal Form (1NF)

Bookstore Order Data

Name	Tayyab Hanif Awan
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Course Title	Database Management System (CS-2204)
Submitted To	Engr. Awais Rathore
Lab No.	04
Tool Used	XAMPP (MySQL / MariaDB) with phpMyAdmin

1. Objective

- Understand the problems caused by an unnormalized table (repeating groups, multi-valued attributes).
- Identify violations of atomicity in a raw bookstore order record.
- Convert the unnormalized order data into First Normal Form (1NF).

2. Database Structure

The database (normalization_lab4) starts from a single unnormalized bookstore order table and decomposes it into the 1NF table OrderDetails_1NF, where every column holds a single, atomic value and repeating book/quantity groups are split into individual rows.

2.1 Table

Table	Purpose
OrderDetails_1NF	One row per order line item — atomic values only, no repeating groups

3. Key Tasks / Concepts Implemented

- Reviewing the original unnormalized order table and identifying which columns violate 1NF.
- Splitting multi-valued 'book list' fields into individual atomic rows.
- Creating and populating the OrderDetails_1NF table with the corrected structure.
- Verifying the 1NF result with a SELECT query.

4. Implementation (phpMyAdmin)

The conversion was implemented and verified using phpMyAdmin under XAMPP.

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-- =====
-- 1. UNNORMALIZED DATA
-- =====
-- Repeating groups are present in the OrderDetails column.

-- Example of UNF:
-- OrderID | CustomerName | CustomerPhone | OrderDetails
-- O101    | Ali Khan    | 0300-1111111 | Book A, Book B, Book C
-- O102    | Sara Ahmed  | 0301-2222222 | Book D

-- Problems in UNF:
-- 1. Repeating groups
-- 2. Difficult searching and updating
-- 3. Data redundancy
-- 4. Possible insertion, update and deletion anomalies
```

Structural overview of unnormalized DB

Server: 127.0.0.1 » Database: normalization_lab4 » Table: orderdetails_1nf

Table structure

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐ 1	OrderID	varchar(10)	utf8mb4_general_ci		No	None			Change Drop More
☐ 2	CustomerName	varchar(100)	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐ 3	CustomerPhone	varchar(20)	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐ 4	BookID	varchar(10)	utf8mb4_general_ci		No	None			Change Drop More
☐ 5	BookTitle	varchar(150)	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐ 6	Quantity	int(11)			Yes	NULL			Change Drop More
☐ 7	Price	decimal(10,2)			Yes	NULL			Change Drop More

Structure tab for implemented 1NF table

Browse Structure SQL Search Insert Export Import Privileges Operations Tra								
☐ Show all Number of rows: 25 Filter rows: Search this table Sort by key: None								
Extra options								
		OrderID	CustomerName	CustomerPhone	BookID	BookTitle	Quantity	Price
☐	Edit Copy Delete	O101	Ali Khan	0300-1111111	B101	Database Systems	1	2500.00
☐	Edit Copy Delete	O101	Ali Khan	0300-1111111	B102	SQL Fundamentals	2	1500.00
☐	Edit Copy Delete	O101	Ali Khan	0300-1111111	B103	Web Development	1	2000.00
☐	Edit Copy Delete	O102	Sara Ahmed	0301-2222222	B104	Python Programming	1	3000.00
☐ Check all With selected: Edit Copy Delete Export								
☐ Show all Number of rows: 25 Filter rows: Search this table Sort by key: None								
Query results operations								

Browse tab showing 1NF data

5. Conclusion

The bookstore order data was successfully converted from an unnormalized form into First Normal Form, removing repeating groups and ensuring every column holds a single atomic value — the foundation for the further normalization done in Lab 5.